

Advanced Microeconomics

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Problem Set 2: Mathematical solutions

2. Application

a) Calculate all partial derivatives and the total derivatives for the following functions:

1)

$$f(x, y) = x^3 y^3$$

Solution:

$$\frac{\partial f}{\partial x} = 3x^2 y^3; \frac{\partial f}{\partial y} = 3x^3 y^2; df = 3x^2 y^3 \cdot dx + 3x^3 y^2 \cdot dy$$

2)

$$f(x, y) = \frac{1}{1 + x^4 + y^2}$$

Solution:

$$\frac{\partial f}{\partial x} = \frac{-4x^3}{(1 + x^4 + y^2)^2}; \frac{\partial f}{\partial y} = \frac{-2y}{(1 + x^4 + y^2)^2}; df = -2 \frac{(2x^3 \cdot dx + y \cdot dy)}{(1 + x^4 + y^2)^2}$$

3)

$$f(x, y) = e^{x^2 y^2}$$

Solution:

$$\frac{\partial f}{\partial x} = e^{x^2 y^2} \cdot 2xy^2; \frac{\partial f}{\partial y} = e^{x^2 y^2} \cdot 2x^2 y; df = 2e^{x^2 y^2} (xy^2 \cdot dx + x^2 y \cdot dy)$$

b) Assume the utility function $U_1 = 4xy^2$.

1) Calculate the MRS.

Solution:

$$MRS_1 = \frac{y}{2x}$$

2) Perform a monotone transformation of U_1 and recalculate the MRS.

Solution:

$$U_2 = (U_1)^2 = 16x^2y^4 = MRS_2 = \frac{32xy^4}{64x^2y^3} = \frac{y}{2x}$$

c) Given is the following utility function $U(x, y) = 4x^2y^2$ and the budget constraint $100 = 8x + 3y$ of a household.

2) What are the optimum quantities of x and y consumed by a household with the given utility function and budget constraint?

Solution:

$$x = 6,25; y = 16,67$$

3) What is the optimum utility of the household?

Solution:

$$U_1 = 43,420.14$$

4) Assume that the price of good x is reduced by 3 monetary units. Calculate the utility of the household in the new optimum.

Solution: $U_2 = 111,155.56$